Pre-Analysis Plan (PAP): The Effects of Having Daughters on Attitudes and Behavior

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1 Introduction

This study investigates the causal impact of having daughters on parental attitudes and behaviors using the German Socio-Economic Panel (SOEP). I focus on how having daughters shapes attitudes toward gender equality, political preferences, and household behavior. My study tests four pre-specified hypotheses using a staggered difference-in-differences (DiD) design to identify the causal effects using observational panel data.

Individuals who have daughters serve as the treated group, and their "treatment time" varies according to the year of the birth of their first daughter. Individuals who never have daughters serve as the untreated comparison group. This design enables the estimation of treatment effects while accounting for temporal variation in treatment timing and unobserved individual heterogeneity.

2 Data

I use the German Socio-Economic Panel (SOEP), a nationally representative longitudinal survey of German households from 1984 onward. The SOEP contains detailed information on respondents' family composition, political attitudes, household dynamics, and gender-role beliefs, making it ideal for this analysis.

I use data from 2000 to 2020 to ensure consistency in variable availability. Observations are at the individual year level.

3 Empirical Strategy: Staggered Difference-in-Differences

3.1 Treatment Definition

- **Treated group**: Parents whose first daughter is born during the observation period.
- **Control group:**: Parents with children but no daughters throughout the observation window.
- **Treatment timing**: The year of the first daughter's birth.

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I estimate treatment effects using a staggered DiD framework. The empirical model of the two way fixed effects regression with event-time indicators is specified as follows:

$$\text{Outcome}_{it} = \sum_{k \neq -1} \beta_k \mathbb{1}_{t-t^*=k} \cdot \text{Treat}_i + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it} \tag{1}$$

Where:

- $Outcome_{it}$ = Outcome variable of interest
- t^* = the year individual i has had a daughter
- $Treat_i$ equals 1 for an individual i that has at least one daughter and 0 otherwise
- α_i = individual fixed effects
- λ_t = year fixed effects
- X_{it} = time-varying individual characteristics (described below)

To account for the possibility of heterogeneous treatment effects and the fact that different individuals have daughters at different times, I follow the methodologies proposed in (Callaway and Sant'Anna, 2021) and (de Chaisemartin and D'Haultfœuille, 2020). Standard errors are clustered at the individual level.

4 Hypotheses

- H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).
  - Dependent Variable: Construct an index of gender-equality attitudes (standardized) based on responses to survey questions on attitudes towards gender
- H2. Having daughters leads to more left-leaning political preferences
  - Dependent Variable: Left-right political inclination scale, higher scale = more left-leaning
- H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
  - Dependent Variable: For respondents living in the same household, calculate the time spent on household chores separately for women and men, and use the proportion of total household chore time contributed by the female respondent as the outcome variable.

    Time on childcare: include both supervised and hands-on care components.
- H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).
  - Estimate equation (1) with support of gender equality as an outcome separately for women and men to obtain the aggregate group-time average treatment effects for each group, and assess whether the effect is larger for men than for women.

5 Covariates and Sample Restrictions

I control for the following time-varying covariates: age, marital status, education level, number of children, household income, employment status, and region (e.g., East or West Germany). Furthermore, to mitigate concerns of self-selection, where individuals who choose to have children may differ systematically from those who do not, I restrict the sample to respondents with at least one biological child. I also exclude respondents under age 18 and restrict the analysis to individuals observed both before and after treatment.

6 Robustness Checks

To assess the robustness of the main results, I re-estimate model (1) using the following alternative definitions of the treatment variable:

1. A dummy indicator that equals one if the respondent's firstborn child is a daughter and zero others.
2. A dummy indicator that equals one if a female child is a later-born daughter. Comparing the results for outcome 1 and 2 can further demonstrate how the effect of having children differ by family structures.
3. The share of daughters among all biological children, which accounts for the proportion of daughters in the family rather than treating child gender as binary.

To further validate the causal interpretation of the results, I conduct placebo tests by replacing the treatment variable with indicators based on the birth of sons. If the original effects are truly driven by having daughters, I should not observe similar patterns when using sons' births as a placebo treatment.

7 Outcome Reporting

Each hypothesis is evaluated using point estimates and 95% confidence intervals. Robust standard errors are clustered at the individual level. I report separate regression tables and event study figures for each hypothesis.

8 Code

Each hypothesis is evaluated in a separate .do file using Stata.

References

Brantly Callaway and Pedro H. C. Sant'Anna. Difference-in-Differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, December 2021. ISSN 0304-4076. doi: 10.1016/j.jeconom.2020.12.001. URL https://www.sciencedirect.com/science/article/pii/S0304407620303948.

Clément de Chaisemartin and Xavier D'Haultfœuille. Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review*, 110(9):2964–2996, September 2020. ISSN 0002-8282. doi: 10.1257/aer.20181169. URL https://www.aeaweb.org/articles?id=10.1257/aer.20181169.